Digital Archives and Methods Portfolio

Portfolio Exercises and Final Project

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Portfolio overview

This portfolio contains four exercises completed during the course before the final project. The exercises show a progression from data cleaning and regular expressions, through basic R functions and CSV import, to text mining in RStudio. The final project develops these skills further by applying word-frequency and sentiment analysis to protest-song lyrics.

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Submission structure

The portfolio is organised as a continuous submission containing four course exercises, the final project report, and the required metadata. The complete R project, data, figures, and script are available in the accompanying GitHub repository.

The permanent GitHub repository is listed in the software metadata section.

Portfolio Element 1 - Regular Expressions and OpenRefine

This exercise introduced regular expressions and data cleaning in OpenRefine.

1. **\b\s?(\d{1,2}).\s?(\d{1,2}).\s?(\d{1,4})** and then substitute using **$3-$2-$1** to get the wanted format of YYYY-MM-DD
2. Firstly, we will wrap each word in quotation marks using the code **^(.+)$**, and then using substitution change the value to "**$1**". Now each word is in quotations. Next using the code **\r?\n** to replace each line break with: **,**. Now each word is in a line with quotations and commas. To turn it back into Voyant style we'll simply find all **"** and substitute them with **blank**. Then we'll find all commas, using the code: **,\s*** and substitute these with line breaks using **\n**
3. No, you're simply changing how the data is displayed by using codes. You must make an active choice to remove or alter the data
4. By cleaning and sorting the data in OpenRefine, the answer is October as the dryest month with 74 reported as months with no water, and September as the second dryest with 70 reported instances
5. I have chosen to look at unmarried women in the age group 20-30. There are in total 2.365 women in this group, and I have excluded the following
    - Sick
    - Not making a living
    - N/A

    The list of 10 most frequent jobs in the dataset
    - Væver - 37 entries
    - Tjenestepige - 28 entries
    - Husholder - 19 entries
    - Inderste - 13 entries
    - Spinder - 13 entries
    - Tjener forældrene - 10 entries
    - I tjeneste - 9 entries
    - Skræder - 7 entries
    - Syerske - 7 entries
    - Køkkenpige - 6 entries

Portfolio Element 2 - Basic R Functions

Week 10 exercise. This task worked with a numeric vector containing missing values and used basic R functions to inspect and summarise it.

```
rooms <- c(1, 5, 2, 3, 1, NA, 3, 1, 3, 2, NA, 1, 8, 3, 1, 4,
           NA, 1, 3, 1, 2, 1, 7, 1, NA, 4, 3, 1, 7, 2, 1, NA,
           1, 1, 3)
sum(rooms>2,na.rm=TRUE)
#According to the code there are 13 elements exceding 2 rooms

typeof(rooms)
#Double

median(rooms,na.rm = TRUE)
#By using na.rm we can exclude the null results,
#which would beforehand be the median.
#The correct answer is 2
```

The exercise uses sum() with a logical condition to count values above two, typeof() to inspect the vector type, and median() with na.rm = TRUE to exclude missing values. The result shows thirteen entries above two and a median of two.

The exercise also clarified the difference between NA and NULL in R. The function median() uses na.rm = TRUE so that missing values do not affect the calculation.

Portfolio Element 3 - CSV Import and Dataset Inspection

This exercise focused on importing a CSV file and inspecting its structure in RStudio.

Danish Kings

1. Assignment

The separator is commas or ,

2. Assignment

Answer 1:

read_cvs() is the tidyverse version

Answer 2:

"spec_tbl_df" "tbl_df" "tbl" "data.frame"

Answer 3:

The dataset has 11 total columns. This is found by using the function ncol(kings)

Answer 4:

	Name	House	Start_year	End_year	Birth_year	Death_year	Gender	Dynasty	Source	BirthDMY	DeathDMY
1	Gorm den Gamle	Gorm	936	958	908	958	M	Jellingdynastiet	https://da.wikipedia.org/wiki/Gorm_den_Gamle	908	958
2	Harald 1. Blåtand	Gorm	NULL	NULL	936	985	M	Jellingdynastiet	https://da.wikipedia.org/wiki/Harald_Bl%C3%A5tand	936	987
3	Toke_Gormsen	Gorm	985	986	NULL	986	M	Jellingdynastiet	https://da.wikipedia.org/wiki/Toke_Gormsen	NA	NA
4	Svend 1. Tveskæg	Gorm	NULL	NULL	NULL	1014	M	Jellingdynastiet	https://da.wikipedia.org/wiki/Svend_Tvesk%C3%A6g	17/04/963	03/02/1014
5	Harald 2.	Gorm	1014	1018	963	NULL	M	Jellingdynastiet	https://da.wikipedia.org/wiki/Harald_2.	994	1018
6	Knud 1. den Store	Gorm	1018	1035	NULL	1035	M	Jellingdynastiet	https://da.wikipedia.org/wiki/Knud_den_Store	995	12/11/1035
7	Hardeknud	Gorm	1035	1042	995	1042	M	Jellingdynastiet	https://da.wikipedia.org/wiki/Hardeknud	1018	08/06/1042
8	Magnus den Gode	Fairhair	1042	1047	1018	1047	M	Jellingdynastiet	https://da.wikipedia.org/wiki/Magnus_den_Gode	1024	25/10/1047
9	Svend 2. Estridsen	Estridsen	1047	1074	1024	1076	M	NA	https://da.wikipedia.org/wiki/Svend_Estridsen	1019	28/04/1076
10	Harald 3. Hen	Estridsen	1074	1080	1019	1080	M	Jellingdynastiet	https://da.wikipedia.org/wiki/Harald_Hen	1041	17/04/1080
11	Knud 2. den Hellige	Estridsen	1080	1086	1041	1086	M	Jellingdynastiet	https://da.wikipedia.org/wiki/Knud_den_Hellige	1043	10/07/1086
12	Oluf 1. Hunger	Estridsen	1086	1095	1043	1095	M	Jellingdynastiet	https://da.wikipedia.org/wiki/Oluf_Hunger	1050	18/08/1095
13	Erik 1. Ejegod	Estridsen	1095	1103	1050	1103	M	Jellingdynastiet	https://da.wikipedia.org/wiki/Erik_Ejegod	1055	10/07/1103
14	Niels	Estridsen	1104	1134	1055	1134	M	Jellingdynastiet	https://da.wikipedia.org/wiki/Niels_af_Danmark	1134	25/06/1134
15	Erik 2. Emune	Estridsen	1134	1137	1065	1137	M	Jellingdynastiet	https://da.wikipedia.org/wiki/Erik_Emune	1100	18/09/1137
16	Erik 3. Lam	Estridsen	1137	1146	1100	1146	M	Jellingdynastiet	https://da.wikipedia.org/wiki/Erik_Lam	1120	27/08/1146
17	Svend 3.; Knud 3.; Valdemar 1.	Estridsen	1146	1157	1120	NA	M	Jellingdynastiet	NA	NA	NA
18	Valdemar 1. den Store	Estridsen	1157	1182	1131	1182	M	Jellingdynastiet	https://da.wikipedia.org/wiki/Valdemar_den_Store	14/01/1131	12/05/1182

Portfolio Element 4 - Text Mining in RStudio

Week 13 exercise. The task used the text of A Game of Thrones to practise PDF extraction, tokenisation, stop-word removal, word counting, and sentiment analysis. The submitted R Markdown file and RStudio project are included in the accompanying portfolio source folder.

```
library(tidyverse)
library(here)
library(pdftools)
library(tidytext)
library(textdata)

got_path <- here("got.pdf")
got_text <- pdf_text(got_path)

got_df <- data.frame(got_text) %>%
  mutate(text_full = str_split(got_text, pattern = "\n")) %>%
  unnest(text_full) %>%
  mutate(text_full = str_trim(text_full))

got_tokens <- got_df %>%
  unnest_tokens(word, text_full)

got_stop <- got_tokens %>%
  anti_join(stop_words)

got_swc <- got_stop %>%
  count(word) %>%
  arrange(-n)
```

The important step in this exercise is the change from a PDF into a structured list of words. Once the text is tokenised, common stop words can be removed and the remaining vocabulary can be counted. This is the same basic workflow later used in the final project, although the final project uses a much smaller and more controlled dataset.

Portfolio Element 4 - Output and Reflection

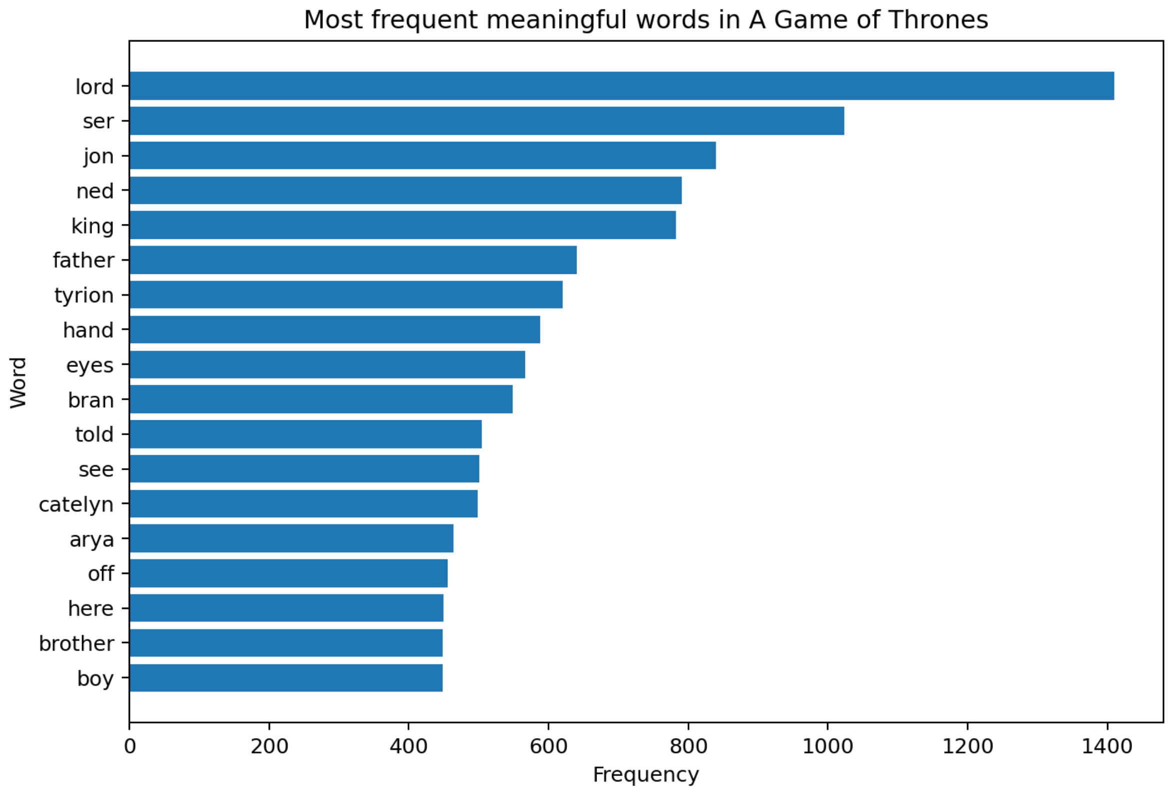

Figure P1. Frequent meaningful words in the submitted A Game of Thrones source after basic stop-word removal.

The output is dominated by character names, titles, and family relationships. This shows why word-frequency results have to be interpreted in relation to the source. A high count does not automatically make a word historically or thematically important; it may instead reflect the structure of a novel or the repeated appearance of major characters.

The exercise also introduced the AFINN sentiment lexicon. This helped prepare the later use of a lexicon-based sentiment method in the final project, while also showing a limitation: assigning a fixed score to individual words cannot fully recognise irony, context, or narrative voice.

Anti-War Sentiment in Popular Music Lyrics During the Vietnam War

Introduction

Music is not only entertainment. It can also be used as a historical source, because lyrics often show what people were concerned about at a specific time. During the 1960s and early 1970s, the Vietnam War became one of the most controversial events in American society. Many young people were drafted into the army, television showed images from the war, and public trust in politicians weakened. In this period, popular music became an important way to express protest, frustration, and political criticism.

This analysis looks at how popular music lyrics reflected anti-war sentiment and political protest during the Vietnam War era. The central research question is: how did protest songs express opposition to war and political authority, and how can digital methods help analyse this development? The project uses twelve songs released between 1962 and 1970. It expects that many of the earlier songs use broader and more symbolic language, while several of the later songs refer more directly to war, soldiers, political leaders, and the unequal effects of the draft.

The goal is not to explain the development of protest music through numbers alone, but to use RStudio as a tool for finding patterns in the lyrics. Word counts and sentiment analysis make it possible to compare all of the selected songs in the same way. At the same time, the results still have to be interpreted historically. A graph can show that a word appears often, but it cannot explain irony, musical performance, or why the word mattered to listeners at the time. This is where close reading and historical context are still necessary.

Sources and Digital Archives

The main primary sources for this project are lyrics from well-known protest songs from the Vietnam War era. The selection contains twelve songs by artists including Bob Dylan, Creedence Clearwater Revival, Edwin Starr, John Lennon, Crosby, Stills, Nash & Young, The Doors, and Country Joe and the Fish. The songs were chosen because they deal with war, peace, political authority, military service, or the wider protest movements of the 1960s. They also represent different ways of protesting, ranging from broad moral questions to direct attacks on political leaders and military inequality.

The lyrics were collected from digital lyric websites such as Genius and AZLyrics, while Billboard was used to investigate chart performance. These websites are not traditional archives in the same way as a state archive, but they still function as digital collections of cultural material. They make lyrics searchable and easy to compare. However, they can also include transcription mistakes, and they often focus on famous artists. Lesser-known protest singers may therefore be missing. The exact webpage used for every lyric was not recorded when the dataset was first created, and the metadata states this limitation rather than assigning uncertain sources afterwards.

Chart information can still help show whether a song reached a broad audience. For example, Billboard records that both “Fortunate Son” and “Give Peace a Chance” reached number 14 on the Hot 100. Chart position does not prove that listeners agreed with the political message, but it does show that the songs circulated beyond small activist groups.[1] [2]

Dataset and Metadata

The dataset contains twelve songs released between 1962 and 1970. Five songs are placed in an early group covering 1962 to 1965, while seven are placed in a later group covering 1968 to 1970. This division makes the comparison manageable, but it should not be understood as a complete history of protest music. The accompanying metadata file records each song's title, artist, year, period, word count, and the available information about where the lyric record came from. This makes the selection easier to inspect and shows which decisions shaped the dataset before the analysis began.

Song	Artist	Year	Group
Where Have All the Flowers Gone	Peter, Paul and Mary	1962	Early
Blowin' in the Wind	Bob Dylan	1963	Early
Masters of War	Bob Dylan	1963	Early
The Times They Are a-Changin'	Bob Dylan	1964	Early
Eve of Destruction	Barry McGuire	1965	Early
The Unknown Soldier	The Doors	1968	Later
Fortunate Son	Creedence Clearwater Revival	1969	Later
Give Peace a Chance	John Lennon	1969	Later
21st Century Schizoid Man	King Crimson	1969	Later
War	Edwin Starr	1970	Later
Ohio	Crosby, Stills, Nash & Young	1970	Later
I-Feel-Like-I'm-Fixin'-to-Die Rag	Country Joe and the Fish	1970	Later

Method: Using RStudio for Text Analysis

The method used is a word-frequency analysis followed by a basic sentiment analysis in RStudio. The lyrics are placed in a CSV file with columns for song title, artist, year, and lyrics. RStudio then splits the lyrics into individual words, removes common stop words, and counts which words appear most often. This makes the analysis more systematic, because it looks across all twelve songs instead of only selecting individual lines that support the argument.

The original CSV uses semicolons as separators. It is therefore imported with read_csv2(), which is the tidyverse version suited to this format. The lyrics are then split into single words with tidytext. Common stop words such as "the", "and", and "is" are removed, together with a short list of vocal fillers such as "yeah", "huh", and "oh". The cleaned words are counted and used to create Figure 1. The Bing lexicon is then used for the sentiment figures.[3]

The full code is kept in the accompanying R file rather than being printed across several pages of the paper. The project folder also contains the dataset, metadata, figures, and a RStudio project file. The script creates the figure and output folders automatically and saves the result tables. Keeping these files together means that another person can follow the same steps and reproduce the figures instead of having to reconstruct the method from screenshots or isolated pieces of code.

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[1] Billboard, "John Lennon Chart History: Hot 100," https://www.billboard.com/artist/john-lennon/chart-history/hsi/ (accessed 4 August 2026).

[2] Billboard, "Creedence Clearwater Revival Chart History: Hot 100," https://www.billboard.com/artist/creedence-clearwater-revival/chart-history/hsi/ (accessed 4 August 2026).

[3] Julia Silge and David Robinson, Text Mining with R: A Tidy Approach (Sebastopol, CA: O'Reilly Media, 2017).

Historical Context

Opposition to the Vietnam War did not develop at one single moment. John Mueller's study of American public opinion found that support was initially relatively high but declined as American casualties increased. The draft made the war personal for young Americans, while television and newspapers made the conflict more visible at home. Research by Paul Burstein and William Freudenburg also shows that public opinion, demonstrations, and the cost of the war became connected to political decision-making in the United States.[4] [5]

The year 1965 is useful as a historical turning point because American involvement changed in scale. The United States had already supported South Vietnam with money, equipment, and military advisers, but large numbers of American combat troops were deployed from 1965, while the bombing campaign also grew. The war therefore became harder to treat as a distant foreign-policy issue. More families were affected by the draft and by growing casualty numbers, and the conflict became part of everyday political debate. This does not mean that protest began in 1965, but it helps explain why later songs could refer more directly to soldiers, military service, political leaders, and events connected to Vietnam.

The division into an early and a later group is therefore meant as an analytical tool rather than a strict border. The early group runs from 1962 to 1965, while the later group begins in 1968 because the selected dataset does not contain songs from 1966 or 1967. This gap is a limitation and means that the comparison cannot show a smooth year-by-year development. Instead, it compares a group of songs written before or around the major escalation with a group written after the war had become longer, more visible, and more controversial. The groups are useful for identifying broader tendencies, but they should not be understood as complete categories into which every protest song would fit.

Music became part of this political environment because it could carry ideas in a form that was emotional, memorable, and easy to repeat. Ron Eyerman and Andrew Jamison argue that music does not only reflect social movements; it can also help create collective identity and keep traditions of protest alive. Michael Kramer similarly connects rock music and the counterculture to debates about citizenship and political participation during the 1960s. These arguments are useful here because the songs in the dataset were not only comments on the war. Several of them were also performed, repeated, and used in situations where people gathered around a shared political message.[6] [7]

Contemporary historical material is especially useful for understanding “Ohio”. The National Archives explains how the expansion of the war into Cambodia intensified anti-war activity and was followed by the killing of four students at Kent State on 4 May 1970. “Ohio” was written as an immediate response to that event. This gives the song a different relationship to history than a broader song such as “Blowin’ in the Wind”, which could be connected to several political causes.

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[4] Paul Burstein and William Freudenburg, “Changing Public Policy: The Impact of Public Opinion, Antiwar Demonstrations, and War Costs on Senate Voting on Vietnam War Motions,” American Journal of Sociology 84, no. 1 (1978): 99–122.

[5] John E. Mueller, “Trends in Popular Support for the Wars in Korea and Vietnam,” American Political Science Review 65, no. 2 (1971): 358–375.

[6] Michael J. Kramer, The Republic of Rock: Music and Citizenship in the Sixties Counterculture (Oxford: Oxford University Press, 2013).

[7] Ron Eyerman and Andrew Jamison, Music and Social Movements: Mobilizing Traditions in the Twentieth Century (Cambridge: Cambridge University Press, 1998).

The growth of mass media also mattered. Radio, television, records, and music magazines helped spread protest songs to a large audience. Political ideas could therefore move through popular culture, not only through speeches, newspapers, or political organisations. This did not mean that every listener agreed with the songs, but it made protest music part of the wider public debate about the war.

This wider circulation also changed the relationship between music and political participation. A song could be heard privately on the radio or a record, but the same chorus could later be repeated at a demonstration or meeting. In that sense, protest music connected individual listening with collective political expression. However, circulation and influence are not the same thing. Chart success can show that a song reached a broad audience, but it cannot tell us whether listeners accepted its argument, ignored the political meaning, or simply liked the music. Contemporary reviews, interviews, and audience surveys would be needed to study that question more closely.

Analysis

Word Frequency

As shown in Figure 1, the word “war” appears much more often than any other word in the dataset. This is not surprising, since war is the main subject connecting the selected songs. Still, the size of the difference needs some explanation. Edwin Starr’s “War” repeats the word throughout its chorus, meaning that one song has a strong effect on the overall count. The figure therefore shows both the importance of war as a theme and a limitation of simple word counting: repeated choruses can make one song influence the full result more than the others.

Other common words include “friend”, “destruction”, “dead”, “soldier”, and “peace”. These words fit the overall subject, but they do not all carry the same meaning. The frequent use of “dead”, “destruction”, and “soldier” shows the connection between the lyrics and themes of violence and military conflict. “Friend” is more difficult to interpret. It may suggest solidarity, but in some songs it is simply part of a repeated line. Looking at the figure is therefore useful for finding patterns, but close reading is needed before deciding what those patterns mean.

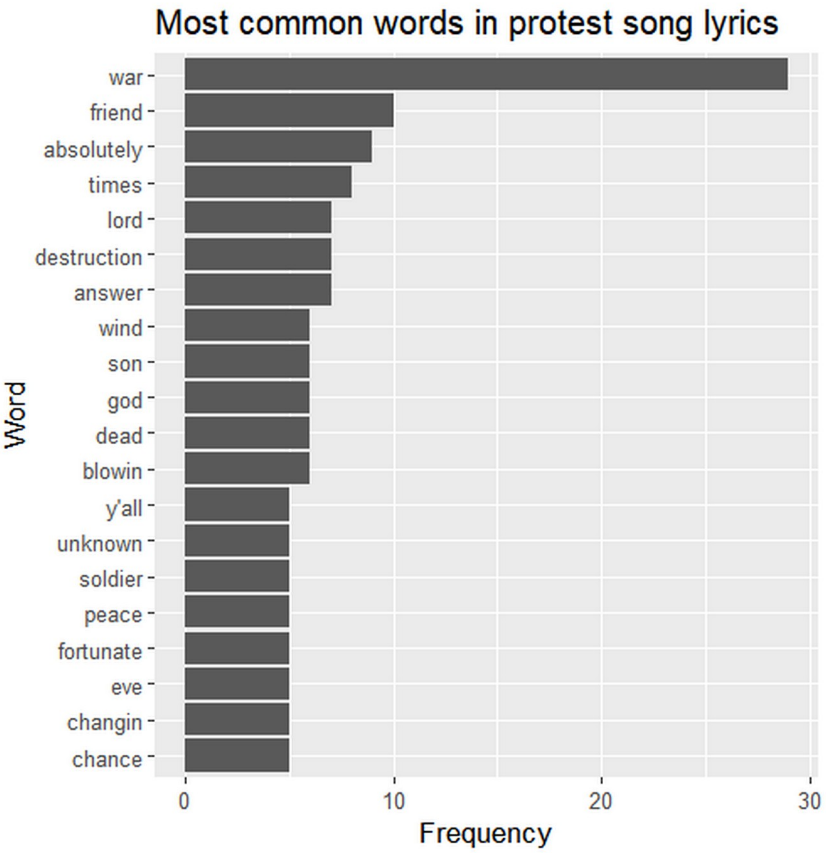


Figure 1. Most common words in the twelve selected songs after stop-word removal.

Early Songs: Moral Questions and Direct Anger

Early protest music often used broad moral questions instead of direct references to Vietnam. “Where Have All the Flowers Gone” presents war as a repeating cycle, while Bob Dylan’s “Blowin’ in the Wind” asks questions about peace, freedom, death, and responsibility. Because the lyrics are open and symbolic, they could be connected to different political movements, including civil rights and anti-war activism. This kind of protest is powerful because it does not depend on one specific event. Instead, it presents war and injustice as wider moral problems.

However, the difference between early and later songs is not completely straightforward. “Masters of War”, released in 1963, is already direct and angry. It attacks those who plan wars while remaining protected from their consequences. This is important because it prevents the paper from making an overly simple claim that early protest was always indirect. Dorian Lynskey’s wider history of protest songs also shows that direct political criticism existed before the major escalation of the Vietnam War.[8]

Later Songs: War, Inequality, and Immediate Events

As the Vietnam War continued, several protest songs became more closely connected to the draft, political leaders, and specific events. Creedence Clearwater Revival’s “Fortunate Son” criticises the unequal burden of military service. The song argues that wealthy and politically connected families could avoid the same risks faced by ordinary people. This makes the song not only anti-war, but also a criticism of class inequality inside the United States. It shows how protest music could connect foreign policy with domestic social problems.

[8]Dorian Lynskey, 33 Revolutions per Minute: A History of Protest Songs (New York: Ecco, 2011).

Edwin Starr's "War" uses a much simpler and more direct strategy. Its central message is repeated until it becomes almost impossible to misunderstand. The song focuses on destroyed lives, grieving families, and the effect of war on a younger generation. "I-Feel-Like-I'm-Fixin'-to-Die Rag" approaches the same subject through dark humour and sarcasm. Together, these songs show that later anti-war music could be direct without always using the same tone.

Crosby, Stills, Nash & Young's "Ohio" is the most event-specific song in the dataset. It names Nixon and refers to the four students killed at Kent State. The lyrics do not give a detailed account of the shootings. Instead, they turn the event into a short and urgent political statement. In this way, the song works almost like a musical newspaper response: it reacts quickly to a contemporary event and makes it part of popular culture.

The songs were divided into two groups to make it easier to compare how protest music changed during the Vietnam War. The first group contains songs released before the large-scale American troop deployment in 1965, while the second group contains songs released after this point. This division was chosen because the role of the United States changed significantly during these years. Before 1965, many protest songs focused on broader themes such as peace, war in general, and moral responsibility. As the war escalated and more American soldiers were sent to Vietnam, the lyrics increasingly became more direct in their criticism of the war, the draft, and political leadership. Looking at the songs in two periods therefore makes it easier to see whether these historical developments are also reflected in the language used.

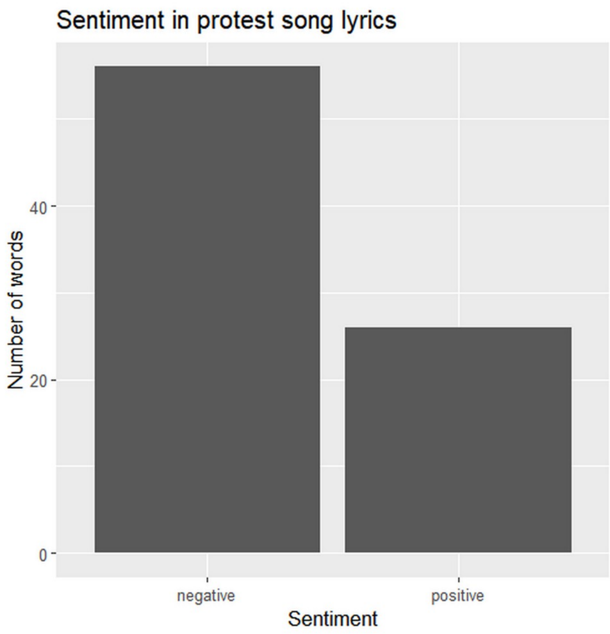
Looking across the two groups, the difference is clearest in how closely several songs attach their criticism to particular political circumstances. The early group often discusses war through repeated cycles, moral questions, or criticism of those who profit from conflict. The later group includes more references to soldiers, the draft, Nixon, and events that listeners could connect directly to the ongoing war. This does not create a perfect chronological progression. "Masters of War" is already extremely direct in 1963, while "Give Peace a Chance" continues to use broad and hopeful language in 1969. The comparison therefore suggests a growing connection to Vietnam and immediate politics rather than a complete replacement of one protest style by another.

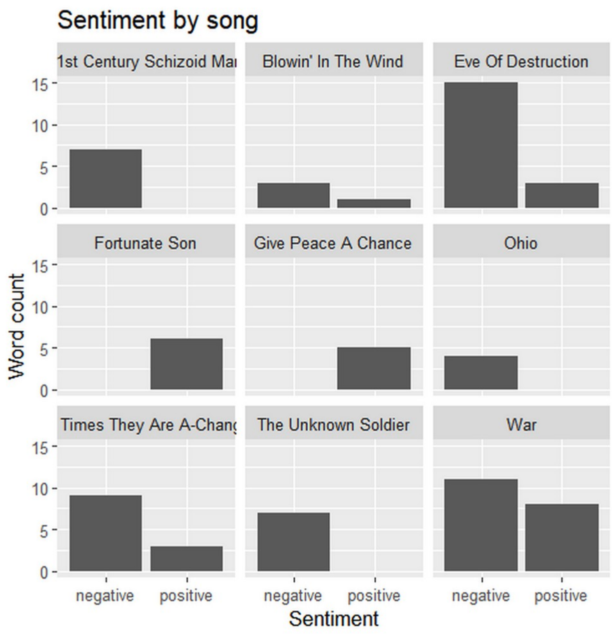
The uneven number and length of songs in each group also affects the comparison. Five songs are placed in the early group and seven in the later group, and songs with repeated choruses contribute more words than shorter songs. For that reason, the groups are mainly used together with close reading rather than as a statistical measurement of all protest music. A larger study could compare equal numbers of songs, normalise frequencies by the total number of words in each group, and test whether the result changes when highly repetitive songs such as "War" are removed.

Sentiment Analysis

The sentiment analysis, shown in Figures 2 and 3, finds more negative than positive words across the selected songs. This fits the general subject of war and political protest. That said, not every song follows exactly the same pattern. "Give Peace a Chance" contains more hopeful and collective language, while songs such as "Eve of Destruction" and "Masters of War" contain a larger number of words classified as negative. Figure 3 is useful because it shows that the overall result is not created equally by every song.

The results still have to be treated carefully. The Bing lexicon assigns individual words a fixed positive or negative value, but it cannot recognise sarcasm, tone, historical context, or the difference between describing violence and supporting it. Repeated choruses also give some words more weight. Sentiment analysis is therefore used as supporting evidence rather than proof that one song or one period was emotionally more negative than another.





Figures 2 and 3. Overall sentiment and sentiment divided by song.

Music as Mobilisation

Protest music did not only reflect the political attitudes of the time; it could also help people express them together. Songs were performed at demonstrations, shared through mass media, and used as tools for collective expression. Eyerman and Jamison's argument about music and collective identity fits "Give Peace a Chance", whose repeated chorus could easily be sung by a crowd. The song does not explain the war in detail, but it gives people a simple and shared language of protest.

Chart success and mass media also mattered because they allowed political messages to move beyond small activist groups. If a protest song appeared on the Billboard chart, this shows that it reached a broader audience. However, popularity should not automatically be treated as evidence that listeners agreed with the message. People could buy or hear a song for many different reasons. Billboard can show circulation, while audience agreement and political effect would require other historical sources, such as contemporary reviews, interviews, letters, or opinion surveys.

Discussion: Strengths and Limits of the Digital Method

The RStudio analysis is useful because it makes the study of lyrics more systematic. It applies the same procedure to every song, identifies repeated vocabulary, and makes it easier to compare several lyrics at once. Saving the steps in a script also means that the analysis can be repeated. This is where digital methods are especially useful: they can reveal patterns that may be difficult to notice when reading twelve songs separately.

This approach also changes how the sources are first approached. A traditional close reading might begin with the most famous lines from one or two songs. The digital method instead begins by treating every lyric in the dataset according to the same rules. That makes the selection of examples less dependent on memory and can draw attention to words or contrasts that deserve further reading. The result is not an automatic explanation, but a way of deciding where historical interpretation should begin.

However, the method also has clear limits. A word-frequency graph cannot explain irony, tone, musical performance, or how an audience reacted. The dataset itself also shapes the result. Twelve well-known songs cannot represent the entire anti-war movement, and the selection is uneven across genre, nationality, race, gender, and political viewpoint. Repeated choruses can make one song dominate a word count. Digital results are therefore shaped by decisions made before the code is run, including which songs were selected, which version of each lyric was used, and which words were removed.

Another limitation is that the method mainly analyses written lyric transcriptions. Protest music also gains meaning through sound, delivery, genre, performance, and the situation in which it is heard. Sarcasm in "I-Feel-Like-I'm-Fixin'-to-Die Rag", for example, is much clearer in performance than in a simple list of words. The analysis therefore works best when it is used to support, rather than replace, close reading and historical source criticism.

The original expectation is only partly supported. Several later songs are more closely connected to Vietnam, the draft, soldiers, and specific events. However, "Masters of War" shows that direct political anger was already present in 1963. The development is therefore not a simple movement from indirect to direct protest. A more careful conclusion is that protest music became more event-specific and more closely connected to the Vietnam conflict as the war intensified, while broader moral criticism continued alongside it.

Conclusion

Popular music was an important part of anti-war protest during the Vietnam War era, but the selected songs show that protest did not take only one form. Some songs use broad questions about peace, freedom, and responsibility. Others attack political privilege, describe the destruction caused by war, or respond directly to events such as the Kent State shootings. The analysis therefore supports the idea that later protest songs often became more closely connected to Vietnam and specific political issues. At the same time, "Masters of War" shows that direct political criticism was already present before the major escalation of the conflict.

Digital methods help make parts of this development visible. Word-frequency analysis shows which terms appear repeatedly across the dataset, while sentiment analysis gives a basic comparison of positive and negative language. These methods do not replace traditional historical analysis. They cannot explain sarcasm, performance, audience reception, or the meaning of a line in its full context. Their main value is that they provide a systematic starting point and make it possible to compare several songs using the same procedure.

The project also shows why documentation matters in digital history. Keeping the lyrics, metadata, R script, figures, and instructions together allows another person to see how the results were produced and to repeat the analysis. The dataset remains limited and mainly represents well-known protest music, but it provides a manageable example of how cultural sources can be studied both as historical texts and as digital data. A larger project could include more songs, lesser-known performers, contemporary reviews, and information about audience reception. This would make it possible to test whether the patterns found here also appear beyond the twelve songs used in this project.

The comparison between the two periods is one of the main lessons from the project. The later songs are often more event-specific and more directly tied to the American experience of Vietnam, but the material does not support a simple story in which symbolic protest suddenly became direct after 1965. Different forms continued alongside each other, and individual artists used very different strategies. This makes protest music useful as a historical source because it shows both changes over time and disagreement within the same political culture.

More broadly, the project shows that digital and traditional historical methods can strengthen each other. RStudio makes repeated vocabulary and broad differences easier to identify, while historical context explains why those differences may matter. Neither approach is sufficient by itself. A larger investigation could add lesser-known performers, women artists, African American protest music, songs supporting the war, and material from outside the United States. It could also compare lyrics with contemporary newspapers, reviews, concert reports, and interviews. That would make it possible to move from the language of protest songs toward a fuller study of how the music was produced, circulated, and understood.

References

Burstein, Paul, and William Freudenburg. 1978. “Changing Public Policy: The Impact of Public Opinion, Antiwar Demonstrations, and War Costs on Senate Voting on Vietnam War Motions.” American Journal of Sociology 84 (1): 99–122.

Eyerman, Ron, and Andrew Jamison. 1998. Music and Social Movements: Mobilizing Traditions in the Twentieth Century. Cambridge: Cambridge University Press.

Kramer, Michael J. 2013. The Republic of Rock: Music and Citizenship in the Sixties Counterculture. Oxford: Oxford University Press.

Lynskey, Dorian. 2011. 33 Revolutions per Minute: A History of Protest Songs. New York: Ecco.

Mueller, John E. 1971. “Trends in Popular Support for the Wars in Korea and Vietnam.” American Political Science Review 65 (2): 358–375.

National Archives. “Episode 9: Crossing into Cambodia.” Remembering Vietnam online exhibition. https://www.archives.gov/exhibits/remembering-vietnam-online-exhibit/episode-9. Accessed 4 August 2026.

Silge, Julia, and David Robinson. 2017. Text Mining with R: A Tidy Approach. Sebastopol, CA: O’Reilly Media.

Billboard. “Creedence Clearwater Revival Chart History: Hot 100” and “John Lennon Chart History: Hot 100.” https://www.billboard.com/artist/creedence-clearwater-revival/chart-history/hsi/ and https://www.billboard.com/artist/john-lennon/chart-history/hsi/. Accessed 4 August 2026.

Required Software and Data Metadata

The following tables document the software, data, and reproducibility information for the final project.

Table 1 - Software metadata

Software metadata description	Project information
Permanent link to GitHub repository	https://github.com/Digital-Methods-HASS/Group-43-Final-Project
Software licence	MIT Licence for the R script and repository documentation.
Data licence	Metadata may be shared under CC BY 4.0. Song lyrics remain copyrighted and are included only for educational analysis; they should not be relicensed as original data.
Operating system and hardware	The analysis was completed on a Windows computer using the desktop versions of R and RStudio.
Software and package versions	R, RStudio, tidyverse, tidytext, and textdata. Running analysis.R saves the exact package and R versions to outputs/session_info.txt.
Installation requirements	R and RStudio, with the packages tidyverse, tidytext, and textdata installed.
Documentation	README.md in the GitHub repository explains the folder structure, required packages, and how to reproduce the figures and output tables.
Support contact	Student contact details are available through Aarhus University and Brightspace.

Table 2 - Data metadata

Metadata description	Project information
data/lyrics.csv	Twelve protest-song records released between 1962 and 1970. Columns include song, artist, year, and lyrics. The semicolon-separated file is imported with read_csv2().
data/metadata.csv	Metadata describing title, artist, year, analytical group, source information, and other recorded fields for each song.
Data acquisition	Lyrics were collected from digital lyric sites including Genius and AZLyrics. Billboard was used for chart context. The exact original webpage for every lyric was not recorded, which is stated as a limitation.
Selection criteria	Songs were selected because they concern war, peace, military service, political authority, or protest movements and provide examples from the early and later parts of the period.
Processing	Lyrics were tokenised into individual words. Standard English stop words and a short list of vocal fillers were removed. Word frequencies and Bing sentiment classifications were then counted and plotted.
Representativeness	The dataset is a small, non-random selection of well-known songs and does not represent all anti-war music, genres, performers, or political viewpoints.
Files produced	Figures are saved in figures/. Count tables and session information are saved in outputs/.

Generative AI declaration

ChatGPT (GPT-5.6 Thinking) was used as a writing and editorial partner during the revision of the portfolio and final project. It supported the organisation of the submission, the setup and review of the GitHub repository, and the improvement of grammar, spelling, clarity, and consistency in text originally written by the student. It was also used to discuss possible revisions and to check whether the submission followed the project requirements. The research topic, source selection, dataset, R analysis, historical interpretation, and final decisions remain the responsibility of the student. All AI-assisted suggestions were reviewed and edited before submission.